

Inelastic Electron Tunneling Spectroscopy at High-Temperatures

Prosper Ngabonziza, Yi Wang, Peter A. van Aken, Joachim Maier, and Jochen Mannhart*

Ion conducting materials are critical components of batteries, fuel cells, and devices such as memristive switches. Analytical tools are therefore sought that allow the behavior of ions in solids to be monitored and analyzed with high spatial resolution and in real time. In principle, inelastic tunneling spectroscopy offers these capabilities. However, as its spectral resolution is limited by thermal softening of the Fermi–Dirac distribution, tunneling spectroscopy is usually constrained to cryogenic temperatures. This constraint would seem to render tunneling spectroscopy useless for studying ions in motion. Here, the first inelastic tunneling spectroscopy studies above room temperature are reported. For these measurements, high-temperature-stable tunnel junctions that incorporate within the tunnel barrier ultrathin layers for efficient proton conduction are developed. By analyzing the vibrational modes of O–H bonds in BaZrO₃-based heterostructures, the detection of protons with a spectral resolution of 20 meV at 400 K (full-width-at-half maximum) is demonstrated. Overturning the hitherto existing prediction for the spectral resolution limit of 186 meV ($5.4 k_B T$) at 400 K, this resolution enables high-temperature tunneling spectroscopy of ion conductors. With these advances, inelastic tunneling spectroscopy constitutes a novel, valuable analytical tool for solid-state ionics.

Inelastic electron tunneling spectroscopy (IETS) is an analytical technique to monitor and analyze accurately the diffusion processes of particles in solids. IETS is highly sensitive and offers temporal resolution plus species specificity. It is used with outstanding success to identify and explore ions and other particles, defects, and elementary excitations in tunnel junctions.^[1–9] By its very nature, IETS is capable of detecting particles in ultrathin barriers or even interfaces. Other spectroscopic techniques often used to characterize ion conductors such as infrared (IR) spectroscopy, Raman spectroscopy, inelastic neutron scattering (INS), and Rutherford backscattering lack this efficiency.^[10–12] Moreover, IETS is sensitive to both IR and

Raman-active transitions. Even the optically forbidden regions that are inaccessible to IR and Raman spectroscopy are open to IETS,^[2,8] making it a promising technique for solid-state ionics. If available, it would open the door to monitoring the simultaneous propagation of one or several particle species in solids separately and in real time. IETS is widely used with great success as a spectroscopy technique for scanning tunneling microscopy.^[5,9,13,14] This variant of IETS is called inelastic scanning tunneling spectroscopy. We focus here on IETS performed by using planar tunnel junctions, which offer multichannel, on-chip integration.

Despite all these attractive properties, IETS has been disregarded as a technique in solid-state ionics to analyze particle motion because the temperatures necessary to study diffusion processes are high, usually well exceeding 300 K. These temperatures were understood to be out of range for IETS. As

early as 1968, J. Lambe and R. C. Jaklevic derived that, due to the thermal softening of the tunnel contacts' Fermi–Dirac distributions, the limit of the spectral resolution of IETS worsens linearly with temperature T , the best possible resolution being $5.4 k_B T$, where k_B is the Boltzmann's constant.^[2] Measurements of the resolution limit of Al–Al₂O₃–Pb junctions between 4.2 and 148 K are consistent with this limit.^[15] Consequently, IETS has been widely considered to be useless above 150 K.^[1–3,16] Room-temperature electron tunneling currents have been employed successfully to identify single amino acids generated by cleaving proteins.^[17] In that work, conductance versus time profiles of the molecules moving through tunnel junctions, were used to identify the molecules, thereby averting the need for spectroscopical resolution. We are aware of only two publications that include room-temperature IETS spectra;^[18,19] indeed room temperature appears to be the highest temperature at which IETS has ever been tried. The temperature gap between IETS and solid-state ionics has also blocked the use of IETS as an analytical tool for nanoionic studies.

However, we reveal here that the limit of $5.4 k_B T$ is not valid at temperatures T above 50 K. At these temperatures, the achievable spectral resolution of IETS can be a factor of 9 better, with no limit found yet. We have obtained a minimal spectral width (full-width-at-half maximum, FWHM) of 19 meV at 300 K and 20.1 meV at 400 K. Consequently, useful IETS can be performed at room temperature and even above,^[19] as we have verified experimentally by our IETS-results relating to protons

Dr. P. Ngabonziza, Dr. Y. Wang, Prof. P. A. van Aken, Prof. J. Maier, Prof. J. Mannhart
 Max Planck Institute for Solid State Research
 Heisenbergstraße 1, Stuttgart 70569, Germany
 E-mail: office-mannhart@fkf.mpg.de

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adma.202007299>.

© 2021 The Authors. Advanced Materials published by Wiley-VCH GmbH. This is an open access article under the terms of the Creative Commons Attribution-NonCommercial License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

DOI: 10.1002/adma.202007299

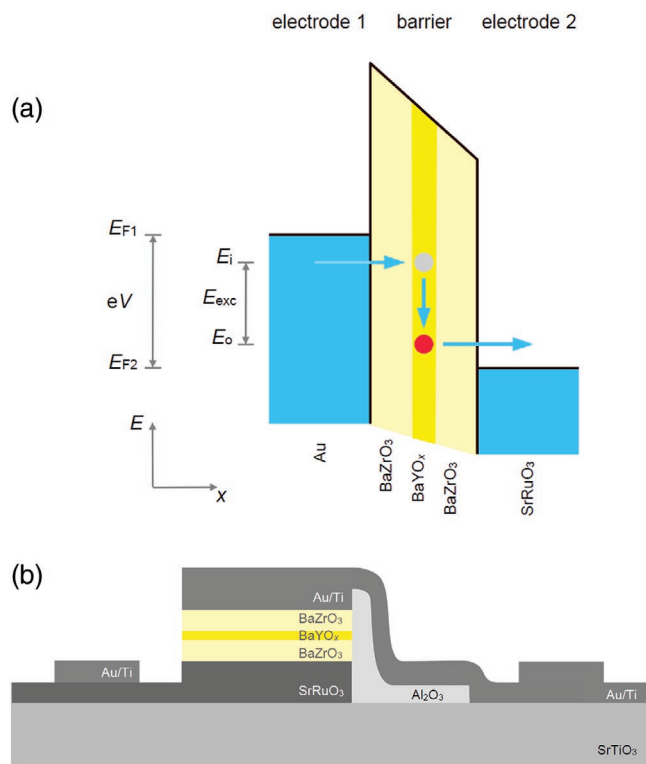


Figure 1. a) Schematic illustration of an inelastic electron tunneling process. Owing to an applied current, an electron leaves electrode 1 and enters the tunnel barrier with an energy of E_i . In the barrier, the electron transfers E_{exc} of its energy to the particle of interest. The electron reaches electrode 2 with an energy of $E_o = E_i - E_{\text{exc}}$. The excitation energy E_{exc} is a characteristic property of the particle of interest, and therefore allows it to be identified and characterized. In this example, the tunneling barrier consists of a BaZrO_3 – BaYO_x – BaZrO_3 trilayer. The embedded BaYO_x layer is a diffusion channel for protons. The energy spectrum of the vibrational modes caused by the protons present in the BaYO_x layer can be derived by measuring E_{exc} . b) Cross section of a tunnel junction according to (a) (not to scale).

diffusing along interfaces in oxide heterostructures up to 400 K. The capability of high-temperature operation proposes IETS as a tool for nanoionics and nanoelectronics.

The operating principle of IETS is illustrated in **Figure 1a**. Electrons with an initial energy E_i are found to tunnel elastically or inelastically across a barrier. In an inelastic tunneling process, a tunneling electron excites a quasiparticle such as a phonon, an ion, an atom, or a molecule. Here, we refer to these species, which may well be non-native to the tunneling barrier, as “particles under study.” An inelastic tunneling process can only occur if E_i exceeds the energy E_{exc} required for the excitation. It is therefore only for $E_i > E_{\text{exc}}$ that inelastic tunneling provides a channel for electrons to cross the barrier. Accordingly, the conductance of the tunnel junction increases stepwise at the tunneling voltage $V = E_{\text{exc}}$, and so the second derivative of the junction’s current–voltage characteristic $\partial^2 I / \partial V^2(V)$ shows a peak. As this peak is centered at E_{exc} , it quantifies the excitation energy and therefore allows the particle under study to be identified. The peak’s strength is a measure of the particles’ concentration and scattering cross sections.

To explore the potential of IETS to study ionic species at high temperatures, we used the same BaZrO_3 -based heterostructures

to act simultaneously as proton conductors and tunnel barriers. BaZrO_3 -based multilayers feature excellent dielectric properties (electronic bandgap 5.33 eV at 300 K), chemical and thermal stability, and high ionic conductivity.^[20–24] Indeed, as proton conductors, such perovskite heterostructures are of great interest for various applications such as solid oxide fuel cells, hydrogen sensors, electrochromic displays, and electrolyzers.^[25–28] Figure 1b shows our fabricated tunnel junctions with Au electrodes grown on top and SrRuO_3 on the bottom. In this architecture, the tunnel barriers provide fast diffusion paths for the particles under study—in our case protons or deuterons—into and through the junctions. The species and the quantity of these particles can then be analyzed in real time by IETS with a spatial resolution defined by the size and pitch of the tunnel junctions.

In the experiments, we used Y-doped BaZrO_3 films. Doping with trivalent acceptor impurities such as Y^{3+} or In^{3+} on the Zr^{4+} site is the common strategy to create oxygen vacancies that enable dissociate water uptake. The water uptake leads to the formation of internal OH^- groups that give rise to proton conductivities in bulk^[29] and thin films^[30] exceeding $10^{-2} \text{ S cm}^{-1}$ at 500 °C (for 20% Y-doping). In uniformly doped $\text{Ba}(\text{Zr}_{1-x}\text{Y}_x)\text{O}_3$ barriers, protons effectively occupy the complete tunnel barrier. This is not always desirable for IETS. To provide a defined local environment for the particles under study and to minimize their interactions with the contacts of the tunnel junctions, we also employed a second approach to enhance proton conduction of BaZrO_3 and take advantage of proton diffusion along interfaces in BaZrO_3 – BaYO_x -based multilayers.^[31] Strong enhancements of ion conduction at interfaces in heterostructures have already been found for several heterostructures, e.g., for CaF_2 – BaF_2 and SrCeO_3 – SrZrO_3 superlattices.^[32,33]

Therefore, the architecture of some of the tunnelling barrier was chosen to take advantage of 2D doping.^[31] The tunnel barrier comprises of a BaYO_x layer embedded between BaZrO_3 layers (Figure 1b). In such a structure, a ZrO_2 layer is replaced in the growth sequence by a YO_x layer, which is intercalated between BaO_x layers, thereby generating oxygen vacancies along the Y-doped layers.^[31] In hydrated conditions, hydroxide ions partially fill these oxygen vacancies, yielding the desired proton-conducting channels located in the middle of the barrier. For reference, we also explored tunnel junctions with barriers consisting of undoped and Y-doped BaZrO_3 layers.

The devices were fabricated by in situ pulsed laser deposition controlled by reflection high-energy electron diffraction and a photolithographic process as described in the Supporting Information. To obtain desirable tunnel resistances, the thicknesses of the BaZrO_3 -based films and heterostructures d were chosen to be between 2.5 and 3 nm, and the tunnel junctions areas A are 10×10 , 20×20 , and $40 \times 40 \mu\text{m}^2$. After the photolithography step, which subjected the samples to water and $\text{C}_4\text{H}_{13}\text{NO}$ -based developer (pH ≈ 13), the samples underwent photoresist baking at 125 °C. To induce further proton (deuteron) diffusion into the samples, several of the tunnel junctions bonded to their carrier chips were soaked in heated regular water or heavy water at later times (see the Supporting Information). Figure 1b and **Figure 2a** show a schematic side view of a device and a top-view scanning electron microscopy image of a tunnel junction, respectively. In total, ten junctions on five substrates were measured: two junctions per substrate for five different substrates labeled A–E.

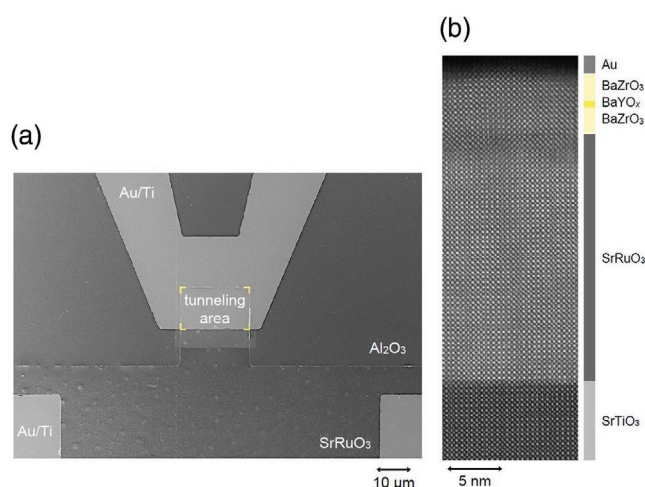


Figure 2. a) Top view scanning electron microscopy image of a typical tunnel junction. The device shown has a proton-conducting BaZrO₃-BaYO_x-BaZrO₃ barrier with an area of 20 × 20 μm². The yellow marks indicate the corners of the tunnel area. b) High-angle annular dark-field STEM image showing a cross-sectional cut through a SrTiO₃-SrRuO₃-BaZrO₃-BaYO_x-BaZrO₃-Au tunnel junction. Epitaxial growth has been maintained through the entire stack, and the thicknesses of the individual layers match the intended device structure. The darkish band in the upper part of the SrRuO₃ layer is attributed to local strain generated by misfit dislocations generated at the interface, as electron energy loss spectroscopy (EELS) O-K edge intensity distribution maps did not show a dip of the oxygen concentration (see Figure S2 in the Supporting Information).

Characterization of the samples by X-ray diffraction and atomic force microscopy reveals a high sample quality (see the Supporting Information). No indications of pinholes were seen. The cross section of a sample was investigated by scanning transmission electron microscopy (STEM). Figure 2b depicts a high-angle annular dark-field-STEM image of a representative SrTiO₃-SrRuO₃-BaZrO₃-BaYO_x-BaZrO₃-Au sample. All layers in the stack are coherent despite the large lattice mismatch between SrRuO₃ ($a = 3.93$ Å) and BaZrO₃ ($a = 4.19$ Å) which induces compressive strain in the BaZrO₃. The barrier thicknesses extracted from STEM data agree well with the X-ray reflectivity measurements (4.8 vs 4.6 nm for the samples of Figure 2b; and Figure S1b (Supporting Information), respectively).

The junctions are highly resistive, and their resistances rise by one order of magnitude if d is enhanced from 2.5 to 3 nm (see Supporting Information). The $I(V)$ characteristics of junction C1 ($d = 2.5$ nm, $A = 20 \times 20$ μm²) are shown in Figure S3b (Supporting Information) for $T = 2$ –400 K. These data were taken on the junction as fabricated, i.e., after photolithography but prior to being soaked in H₂O or D₂O. The maximal voltage range of ± 650 mV was chosen to prevent electric breakdown and excessive heating. The curves are nominally symmetric and overlap for $T \leq 150$ K. At higher T , the conductance of the junctions increases in accordance with a growing contribution of thermally activated electron transport. These properties match the behavior expected for electron transport by tunneling plus thermally activated electron hopping over the barrier.

Having established that the devices act as tunnel junctions, we now present the IETS spectra. IETS measurements were

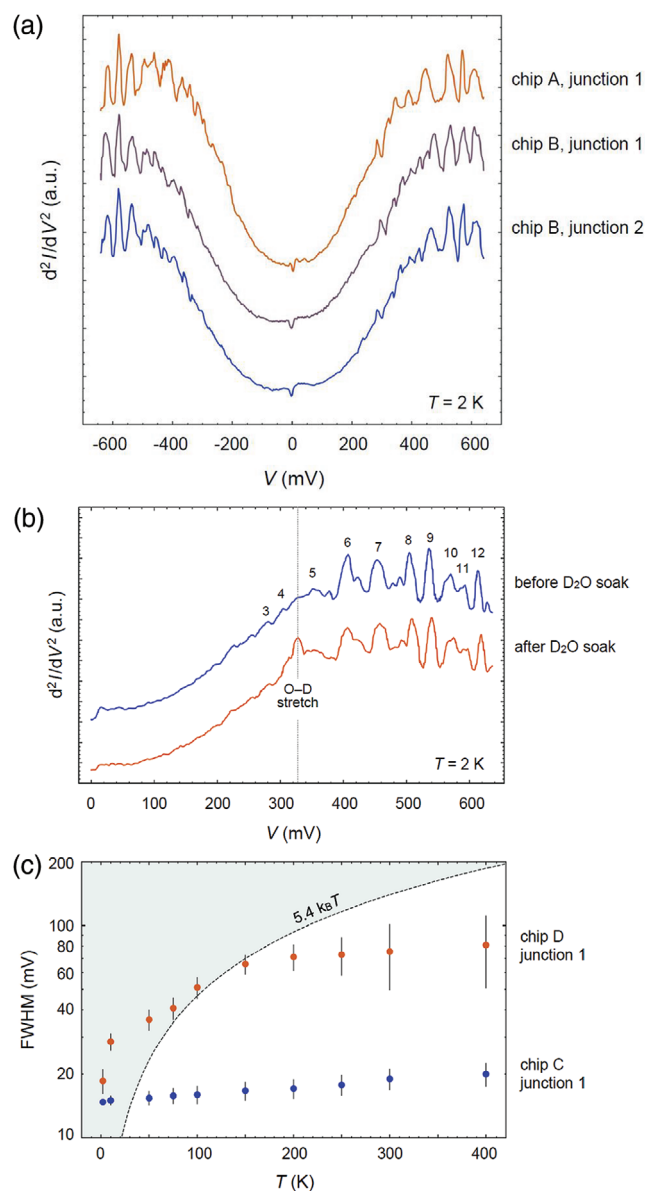


Figure 3. a) IETS spectra of three 2.5 nm thick SrTiO₃-SrRuO₃-BaZrO₃-BaYO_x-BaZrO₃-Au tunnel junctions measured at 2 K. The positive bias corresponds to electrons tunneling into the Au top contact. Chips A and B were subject to different deposition runs. For clarity, the curves have been shifted along the y-axis. b) Comparison of the IETS spectra of a SrTiO₃-SrRuO₃-BaZrO₃-BaYO_x-BaZrO₃-Au tunnel junction (sample B3) before (blue) and after (orange) being soaked in D₂O. The 328 meV peak generated by the D₂O soak reflects the O-D stretching mode. For clarity, the curves have been shifted along the y-axis. c) FWHM peak width of the 580 meV mode (peak 11 in Figure 4c) for two samples C1 and D1 as a function of temperature. Junction D1 is the junction with the broadest linewidth of all. Peak widths outside the green area are below the Lamb-Jaklevic limit of 5.4 $k_B T$ (hatched line).

performed between 2 and 400 K. The upper limit was chosen by wiring-induced constraints. The $\partial^2 I / \partial V^2(V)$ characteristics were directly measured using a standard 4-terminal lock-in-amplifier technique that recorded the second harmonic response of the tunnel junctions (see Supporting Information). The

characteristics were reproducible, also with respect to different deposition runs on different SrTiO₃ substrates (Figure 3a) and were stable for at least 6 weeks. Small voltage differences, at which characteristic features occur among different samples, are attributed to the 4-point geometry being not perfect and, possibly, to minor sample heating.

The $\partial^2 I / \partial V^2(V)$ characteristic of sample C1 is shown in Figure 4. In forward (positive) bias, electrons tunnel from the bottom SrRuO₃ film to the Au top contact. The $\partial^2 I / \partial V^2(V)$ curve shows a rich spectrum of features, which we found reproducibly in all junctions with barriers consisting of Ba(Zr_{0.8}Y_{0.2})O₃ films or of (BaZrO₃)₂–(BaYO_x)₁–(BaZrO₃)₂ multilayers. The details of these features are also independent of deposition run, SrTiO₃ substrate, and scan direction (see Figure 3a; and Supporting Information). However, tunnel junctions with undoped BaZrO₃ barriers showed only a minute peak structure (see Supporting Information). As the proton conductivity is much lower in BaZrO₃ than in Ba(Zr_{0.8}Y_{0.2})O₃ or BaZrO₃–BaYO_x–BaZrO₃,^[20,22,23,24,29,30,31] this observation suggests that the observed peaks are associated with the presence of protons in the barrier.

The spectra reveal three ranges of energy with distinct characteristics: first, a low-energy region (<60 mV) with several small but sharp peaks of $\partial^2 I / \partial V^2(V)$ (Figure 4b), second, an energy window from 60 to 230 mV that is nominally devoid of IETS structures (Figure 4a) and, third, the energy range above 230 mV, which features a rich spectrum of large, crisp peaks (Figure 4c). We now analyze the peaks found in the low and high-energy ranges.

As Figure 4b shows, the height of the peaks in this energy range is partly asymmetric with respect to bias direction, whereas the peak energies are symmetric. The asymmetry in the peak height is common for inelastic tunneling processes. The intensity of a peak in an IETS spectrum is high if the inelastic interaction occurs close to the electrode via which the electrons leave the tunneling barrier,^[4,34] which for the spectra shown in Figure 4b is the Au contact. The energy of these peaks is in the range of tens of meV, which is the range of phonon energies. Indeed, the two main peaks (15 and 53 mV) labeled 1 and 2 in Figure 4b, match optical phonon modes (14.5 and 55 meV) measured by IR and Raman spectroscopy.^[11,35,36] In BaZrO₃, the Ba–ZrO₆ stretching mode (in our case also the

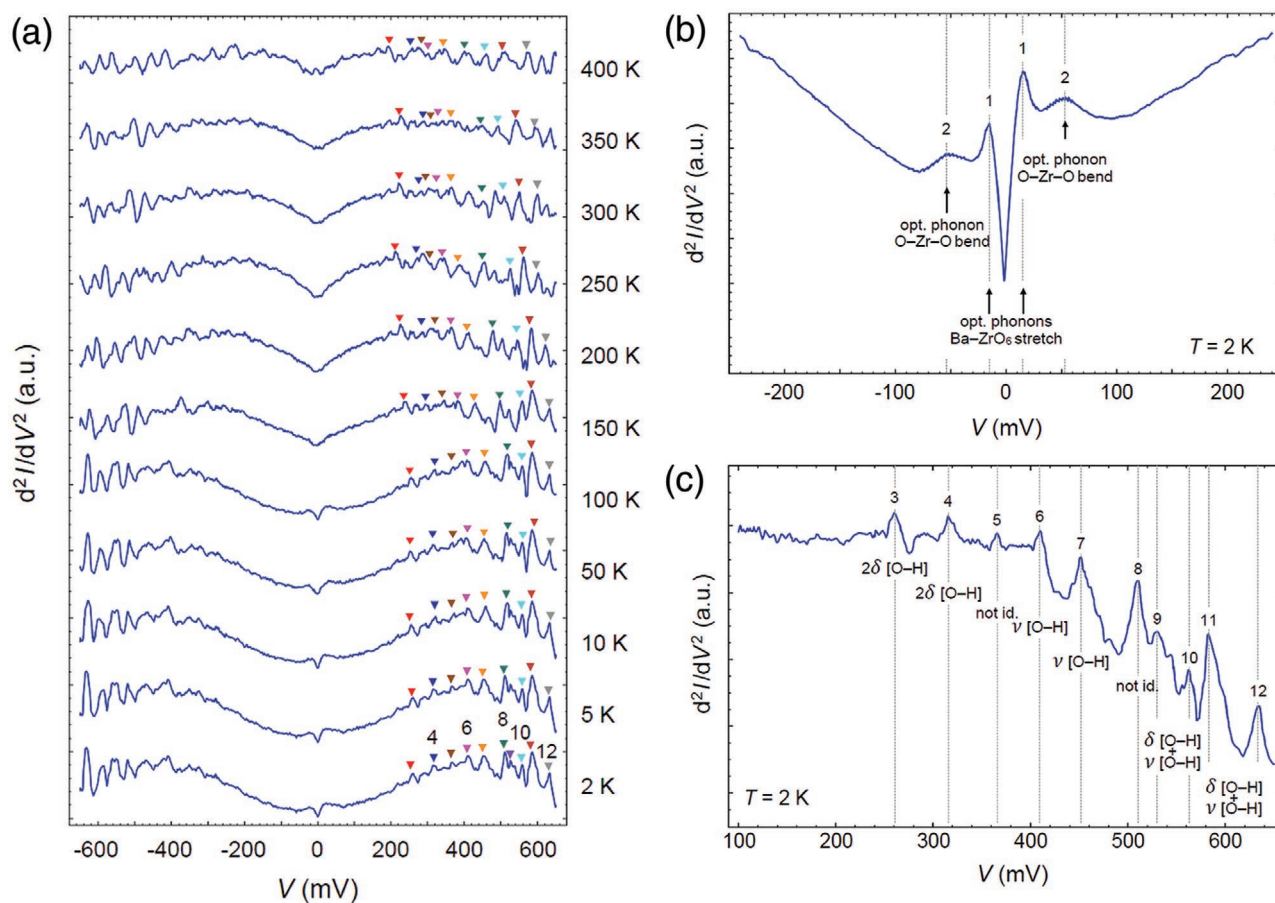


Figure 4. a) IETS spectra of sample C1 measured at a range of temperatures. The low-energy IETS features (<60 meV, peaks 1 and 2) weaken and soften with increasing temperature, giving way to a rounded dip. The high-energy features (>230 meV) are well-resolved and traced up to 400 K, the highest temperature used. The curves have been shifted along the y-axis for clarity. b) Low-energy IETS spectrum of sample C1 measured at 2 K revealing peaks 1 and 2 (15 and 53 meV) resulting from two optical phonon modes in BaZrO₃. The V-shape dip at $V \approx 0$ has only been observed in samples containing Y (see the Supporting Information). c) High-energy IETS spectrum of sample C1 measured at 2 K displaying the assigned vibrational modes. Peaks 5 and 8 have not been identified.

Ba–YO₆ stretching) is known to generate a vibrational band at 15 meV, whereas the O–Zr–O bending mode induces a band at 53 meV.^[11,36] As expected for phonon-assisted tunneling, the energies of the modes remain unchanged with increasing temperature, and the width of the peaks increases (Figure S6, Supporting Information). The peaks start to merge above 10 K.

In the high-energy range, the IETS spectrum resolves ten pronounced, reproducible peaks (Figure 4c). A comparison with reported IR spectra, INS spectra, and results of ab initio molecular dynamics calculations allows us to attribute most peaks to vibrational modes involving hydrogen or to their higher-order transitions.^[36–39] Hydrogen is supposed to be absent in the idealized barrier structure of our samples, but hydration causing proton, hydrogen, hydroxyl, or water diffusion is bound to occur during the photolithography steps and subsequent sample heating. Most peaks observed in Figure 4c match vibrational modes in proton-conducting oxides.^[10,36,38] We use the labeling scheme of Figure 4c in this discussion.

The 2 δ [O–H] higher-order transitions of the O–H bend mode have been found by INS to occur in In-doped BaZrO₃ at 260 and 315 meV. These energies precisely match peaks 3 and 4 of the IETS spectrum.^[37] Peak 7 at 450 meV is a peak that has been prominent in the IETS spectra of a variety of metal–oxide–metal tunnel junctions since the 1960s.^[1,2,18,40] It has been attributed to a ν [O–H] mode, with free OH[–] being generated by oxidation processes in the presence of water and subsequent redistribution of hydrogen bonds. As reported in Ref. ^[37], this 450 meV ν [O–H] mode is usually redshifted in hydrated oxides. Peak 6 (410 meV) agrees well with this redshifted mode so that we tentatively attribute it to the ν [O–H] mode arising from hydration in the oxide barrier. Hydration of the junctions is to be expected, as described above. Furthermore, peaks arising from higher-order transitions and combinations of the δ [O–H] and the ν [O–H] modes are anticipated.^[37] Indeed, peaks 9–12 at 525, 560, 580, and 635 meV, respectively, exactly match the expected energies of the δ [O–H]+ ν [O–H] and 2 δ [O–H]+ ν [O–H] modes (Figure 4c). We cannot assign the remaining peaks 5 (365 meV) and 8 (510 meV) to any combination of δ [O–H] and ν [O–H] modes reported for acceptor-doped BaZrO₃. We therefore suggest that these peaks indicate modes associated with oxygen vacancies or related defects present in the barrier.

It is noteworthy that some of the known O–H bending or stretching modes are weak or even absent from the spectrum. Such a behavior of IETS spectra was already noted and analyzed in the early days of IETS.^[41,42] Although no strict selection rules exist for IETS, propensity rules for the peak strengths have been defined that take into account the effects at the position, orientation, and screening of the dipole electric field of the modes interacting with the tunneling electrons.^[41–46] In addition, the spectra of our junctions are characterized by the almost featureless energy range from 60 to 230 mV. The peaks expected in this range are muted by an unknown phenomenon.

Isotope replacement is a control experiment capable of revealing the role of protons in vibrational modes. We therefore inserted deuterons into the barriers by subjecting the tunnel junctions to D₂O (see Supporting Information). Figure 3b shows the $\partial^2 I / \partial V^2(V)$ characteristics of a SrTiO₃–SrRuO₃–BaZrO₃–BaYO_x–BaZrO₃–Au tunnel junction (B3, $d = 2.5$ nm) before and after exposure to deuterium, which induces the marked

peak at 328 mV that matches the O–D stretching mode.^[47,48] The same peak was reported for deuterated Al–Al₂O₃–Pb junctions as being the only peak above 230 mV.^[2] Its presence confirms that deuterons have penetrated the tunnel barrier and thereby provides evidence that i) protons indeed readily diffuse into the barrier and ii) their vibrational modes generate peaks in the high-energy range of the $\partial^2 I / \partial V^2(V)$ characteristics.

Although one might expect that during the D₂O-soaking experiment some of the protons in the OH-groups to be exchanged with deuterons, the IETS spectra reveal no noticeable shifts of the O–H peaks. It seems that the D₂O-soaking parameters used did not lead to a general replacement of the protons embedded in the crystal lattices. These results suggest that it is much easier to add protons during sample fabrication than to replace them afterward.

IETS could be highly useful if it were possible to perform tunneling spectroscopy above room temperature. This requires a spectral resolution sufficient to yield a peak structure in the $\partial^2 I / \partial V^2(V)$ characteristics that allows the modes to be identified and possibly quantified. Figure 4a displays high-resolution IETS spectra of sample C1 taken between 2 and 400 K that clearly show temperature-driven changes in the characteristics. Importantly, however, the spectra show well-resolved sets of peaks at all temperatures, even at 400 K. Many of these peaks are connected to the previously identified low-temperature modes (Figure 4c). The intensity of the phonon peaks (peaks 1 and 2) weakens with rising temperature. Above 100 K, the sharp zero-bias V-shaped depression of the $\partial^2 I / \partial V^2(V)$ characteristic smoothens into a dip. This behavior is expected from the thermal softening of the electrodes' Fermi–Dirac distributions.^[8,16]

Above 200 meV, the measured IETS peaks, which reflect the O–H vibration bands, persist up to 400 K (Figure 4a). Starting at 150 K, the spectra show that the O–H modes shift by ≈ 10 mV with temperature increasing to 400 K (for peak 3, e.g., by 15 mV). The mechanism causing the observed peak shift, which at its face value would imply a thermal softening of the O–H stretching modes, remains to be investigated.

The peak widths of the 450 meV ν [O–H] modes (15.2 meV) at low temperatures are much smaller than the corresponding peak widths obtained by IR and Raman measurements on polycrystalline BaZrO₃-based bulk samples, but are comparable to the IETS peak widths obtained at Al–Al₂O₃–Pb junctions (≈ 17 meV).^[2,48] It is reasonable to expect that the peak widths of our samples are smaller than those of the bulk polycrystals because the epitaxially oriented, few-unit-cell thick tunnel barriers provide a more regular microstructure.

We did not expect that the IETS spectra would exhibit spectral peaks that are sufficiently distinct at 400 K to identify the presence of protons at interfaces. Indeed, this finding disagrees with the Lambe–Jaklevic limit of the temperature-dependent spectral resolution of $5.4 k_B T$ for the minimal peak width (FWHM). As Figure 3c shows, the measured peak widths, determined as described in the Supporting Information, indeed violate this limit above a sample-specific temperature of 30–150 K. This is shown most pronouncedly by sample C1. For this sample, the width of the 580 meV mode (peak 11) equals 20.1 meV at 400 K, which is a factor of 9 smaller than the $5.4 k_B T$ limit of ≈ 186 meV. At 300 K, the peak width equals ≈ 19 meV, corresponding to $0.75 k_B T$. Such peak widths were found in

most—but not all—of the samples investigated. For example, the peak widths of sample D1 are larger (75.5 and 81 meV at 300 and 400 K, respectively, see Figure 3c), but still less than half of the Lambe–Jaklevic limit. We attribute the observed resolution enhancement to a spectral sharpening of the tunneling electrons induced by sample-dependent resonant states. In this hypothetical scenario, the standard energy distribution of the tunneling electrons is sharpened by resonant electron tunneling, which occurs in addition to direct tunneling, in case the resonant states have a spectral width that is much smaller than the thermal energy distribution of the electrons in the electrodes (see Figure S7 in the Supporting Information). This resonant tunneling is reminiscent to the previously reported orbital-mediated tunneling through molecules.^[49,50] In our case, defects of the oxide-crystal lattices or interface states may provide for the resonant states. With this high resolution, IETS has the capability to yield valuable spectral information even at temperatures above 400 K. To achieve such high operation temperatures requires further optimization of the junctions' thermal stability, including their contacts and wiring.

Having demonstrated that IETS with planar tunnel junctions is a viable high-resolution technique to analyze mobile species in solids, we draw attention to the fact that this technique is only useful to perform spectroscopy on solids that can be integrated in tunneling barriers. This implies that the materials must be excellent insulators, even at layer thicknesses as small as several nanometers. There is no explicit need for the layers to be grown with epitaxial order. The requirement to embed the materials of interest in tunnel junctions implies that IETS is not a suitable technique to study test samples of random bulk materials. Nevertheless, we foresee IETS as a highly valuable tool to analyze materials and heterostructures grown as thin layers specifically for the task of exploring their ionic dynamics. In this case, lithography with all its available freedom for design

can be used to design experimental setups. Figure 5 shows a corresponding, envisaged nanoionic device that utilizes nanostructured tunnel junctions for time resolved, 2D mapping of ionic diffusion in heterostructure devices.

In conclusion, we have designed and validated tunnel junctions in which the tunnel barrier consists of efficient proton conductors. The barriers of these junctions therefore have a double function. They act as a vertical tunneling barrier for electrons and as a horizontal diffusion path for ions. The tunnel junctions yield meaningful IETS spectra of diffused ions up to at least 400 K with no upper bound identified. The excellent fit of the peak energies in the IETS spectra enables them to be attributed to established phonon modes and to O–H bend and stretch modes, which has been confirmed by spectroscopy of tunnel junctions subjected to D₂O.

This work provides evidence that, despite a well-accepted tenet to the contrary, IETS is a powerful spectroscopic tool, capable to analyze ions above room temperature. At 400 K, the achieved spectral resolution of the junctions is at least a factor of 9 better than the established theoretical limit. Applying IETS to ionic conductors opens the door to studying diffusion in solids in real time with a spatial resolution limited only by the minimum size of the tunnel junctions. We expect future IETS to utilize even smaller tunnel junctions, including junctions fabricated from 2D van der Waals materials, to operate at even higher temperatures, and to monitor the motion of several ionic species, such as protons, lithium, and sodium ions simultaneously. With the capability of high-temperature operation, IETS has the potential to become a valuable item in the suite of analytical tools for future solid-state ionics.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors acknowledge valuable discussions in particular with R. Merkle, H. Boschker, D. Braak, P. Bredol, S. Kivelson, T. Harada, and T. Whittles. They also thank M. Hagel and H. Hoier for technical support, and L.-M. Pavka for editorial help. Y. Wang and P. A. van Aken acknowledge funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 823717-ESTEEM3.

Open access funding enabled and organized by Projekt DEAL.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

high-temperature tunneling spectroscopy, inelastic tunneling spectroscopy, ionic diffusion, nanoionics, tunnel junctions

Received: October 25, 2020

Revised: December 4, 2020

Published online: January 18, 2021

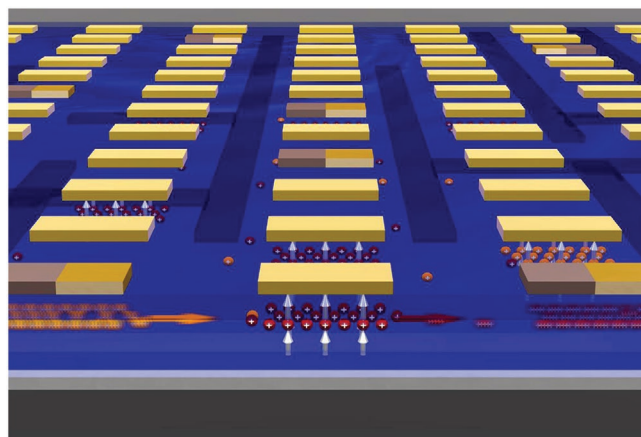


Figure 5. Illustration of a nanoionic device with tunnel sensors monitoring the diffusion of two species of ions (orange and red spheres) along ion-conducting passages (bright blue). Ions diffusing in plane are detected by transversely tunneling electrons (silverish arrows). The ions induce inelastic electron tunneling, which causes ion-specific features in the electron tunneling characteristic. These can be monitored and spatially resolved in real time. The bicolored brown bars depict electrodes. Electric voltages applied between their two halves and voltages applied between two halves of a bottom electrode (not shown) pump the ions in plane.

- [1] R. C. Jaklevic, J. Lambe, *Phys. Rev. Lett.* **1966**, 17, 1139.
- [2] J. Lambe, R. C. Jaklevic, *Phys. Rev.* **1968**, 165, 821.
- [3] S. K. Khanna, J. Lambe, *Science* **1983**, 220, 1345.
- [4] J. W. Reiner, S. Cui, Z. Liu, M. Wang, C. H. Ahn, T. P. Ma, *Adv. Mater.* **2010**, 22, 2962.
- [5] S. You, J.-T. Lü, J. Guo, Y. Jiang, *Adv. Phys. X* **2017**, 2, 907.
- [6] C. J. Adkins, W. A. Phillips, *J. Phys. C: Solid State Phys.* **1985**, 18, 1313.
- [7] M. A. Reed, *Mater. Today* **2008**, 11, 46.
- [8] K. W. Hipps, U. Mazur, in *Tunneling Spectroscopy* (Ed: P. K. Hansma), Springer, Boston, MA **1982**, pp. 229–269.
- [9] E. L. Wolf, *Principles of Electron Tunneling Spectroscopy*, 2nd ed., Oxford University Press, Oxford **2012**.
- [10] A. Perrichon, M. Jiménez-Ruiz, L. Mazzei, S. M. H. Rahman, M. Karlsson, *J. Mater. Chem. A* **2019**, 7, 17626.
- [11] F. Giannici, M. Shirpour, A. Longo, A. Martorana, R. Merkle, J. Maier, *Chem. Mater.* **2011**, 23, 2994.
- [12] L. Mazzei, M. Wolff, D. Pergolesi, J. A. Dura, L. Börjesson, P. Gutfreund, M. Bettinelli, T. Lippert, M. Karlsson, *J. Phys. Chem. C* **2016**, 120, 28415.
- [13] A.-S. Hallböck, N. Oncel, J. Huskens, H. J. W. Zandvliet, B. Poelsema, *Nano Lett.* **2004**, 4, 2393.
- [14] L. J. Lauhon, W. Ho, *Phys. Rev. B* **1999**, 60, R8525.
- [15] R. J. Jennings, J. R. Merrill, *J. Phys. Chem. Solids* **1972**, 33, 1261.
- [16] P. K. Hansma, in *Tunneling Spectroscopy* (Ed: P. K. Hansma), Springer, Boston, MA **1982**, p. 1.
- [17] T. Ohshiro, M. Tsutsui, K. Yokota, M. Furuhashi, M. Taniguchi, T. Kawai, *Nat. Nanotechnol.* **2014**, 9, 835.
- [18] J. S. Moodera, J. Nowak, R. J. M. van de Veerdonk, *Phys. Rev. Lett.* **1998**, 80, 2941.
- [19] V. Bommisetty, S. Bhandari, R. L. Karmacharya, D. A. Rislov, R. D. Mileham, D. Galipeau, D. W. Galipeau, *IEEE Sens. J.* **2008**, 8, 983; Note: Figure 6 in this reference shows a characteristic of a tunnel junction with a 300 K FWHM of ≈ 136 meV $\approx 5.3 k_B T$, and the authors of this reference point out that this width is smaller than the $5.4 k_B T$ limit.
- [20] K. D. Kreuer, *Annu. Rev. Mater. Res.* **2003**, 33, 333.
- [21] E. Fabbri, L. Bi, D. Pergolesi, E. Traversa, *Adv. Mater.* **2012**, 24, 195.
- [22] J. H. Shim, T. M. Gür, F. B. Prinz, *Appl. Phys. Lett.* **2008**, 92, 253115.
- [23] D. Pergolesi, E. Fabbri, A. D'Epifanio, E. Di Bartolomeo, A. Tebano, S. Sanna, S. Licoccia, G. Balestrino, E. Traversa, *Nat. Mater.* **2010**, 9, 846.
- [24] A. Fluri, A. Marcolongo, V. Roddatis, A. Wokaun, D. Pergolesi, N. Marzari, T. Lippert, *Adv. Sci.* **2017**, 4, 1700467.
- [25] H. Malerød-Fjeld, D. Clark, I. Yuste-Tirados, R. Zanón, D. Catalán-Martínez, D. Beeaff, S. H. Morejudo, P. K. Vestre, T. Norby, R. Haugsrud, J. M. Serra, C. Kjølsseth, *Nat. Energy* **2017**, 2, 923.
- [26] S. H. Morejudo, R. Zanón, S. Escolástico, I. Yuste-Tirados, H. Malerød-Fjeld, P. K. Vestre, W. G. Coors, A. Martínez, T. Norby, J. M. Serra, C. Kjølsseth, *Science* **2016**, 353, 563.
- [27] J. A. Kilner, M. Burriel, *Annu. Rev. Mater. Res.* **2014**, 44, 365.
- [28] L. Malavasi, C. A. J. Fisher, M. S. Islam, *Chem. Soc. Rev.* **2010**, 39, 4370.
- [29] K. D. Kreuer, *Solid State Ionics* **1999**, 125, 285.
- [30] Y. B. Kim, T. M. Gür, H.-J. Jung, S. Kang, R. Sinclair, F. B. Prinz, *Solid State Ionics* **2011**, 198, 39.
- [31] P. Ngabonziza, R. Merkle, Y. Wang, P. van Aken, T. S. Bjørheim, J. Maier, J. Mannhart, *Adv. Energy Mater.* **2020**, 2003267, <https://doi.org/10.1002/aenm.202003267>.
- [32] N. Sata, K. Eberman, K. Eberl, J. Maier, *Nature* **2000**, 408, 946.
- [33] H. Yugami, H. Naito, H. Arashi, *Appl. Surf. Sci.* **1997**, 113–114, 222.
- [34] W. He, T. P. Ma, *Appl. Phys. Lett.* **2003**, 83, 2605.
- [35] C. H. Perry, D. J. McCarthy, G. Rupprecht, *Phys. Rev.* **1965**, 138, A1537.
- [36] M. Karlsson, I. Ahmed, A. Matic, S. G. Eriksson, *Solid State Ionics* **2010**, 181, 126.
- [37] L. Mazzei, A. Perrichon, A. Mancini, G. Wahnström, L. Malavasi, S. F. Parker, L. Börjesson, M. Karlsson, *J. Mater. Chem. A* **2019**, 7, 7360.
- [38] M. Karlsson, M. E. Björketun, P. G. Sundell, A. Matic, G. Wahnström, D. Engberg, L. Börjesson, I. Ahmed, S. Eriksson, P. Berastegui, *Phys. Rev. B* **2005**, 72, 094303.
- [39] T. Omata, M. Takagi, S. Otsuka-Yao-Matsuo, *Solid State Ionics* **2004**, 168, 99.
- [40] M. C. Gallagher, Y. B. Ning, J. G. Adler, *Phys. Rev. B* **1987**, 36, 6651.
- [41] J. Kirtley, D. J. Scalapino, P. K. Hansma, *Phys. Rev. B* **1976**, 14, 3177.
- [42] D. J. Scalapino, S. M. Marcus, *Phys. Rev. Lett.* **1967**, 18, 459.
- [43] A. Gagliardi, G. C. Solomon, A. Pecchia, T. Frauenheim, A. Di Carlo, N. S. Hush, J. R. Reimers, *Phys. Rev. B* **2007**, 75, 174306.
- [44] J. Kirtley, J. T. Hall, *Phys. Rev. B* **1980**, 22, 848.
- [45] N. Lorente, M. Persson, L. J. Lauhon, W. Ho, *Phys. Rev. Lett.* **2001**, 86, 2593.
- [46] A. Troisi, M. A. Ratner, *Nano Lett.* **2006**, 6, 1784.
- [47] J. Igalson, J. G. Adler, *Phys. Rev. B* **1983**, 28, 4970.
- [48] R. C. Jaklevic, M. R. Gaerttner, *Appl. Phys. Lett.* **1977**, 30, 646.
- [49] U. Mazur, K. W. Hipps, *J. Phys. Chem. B* **1999**, 103, 9721.
- [50] L. Scudiero, D. E. Barlow, U. Mazur, K. W. Hipps, *J. Am. Chem. Soc.* **2001**, 123, 4073.